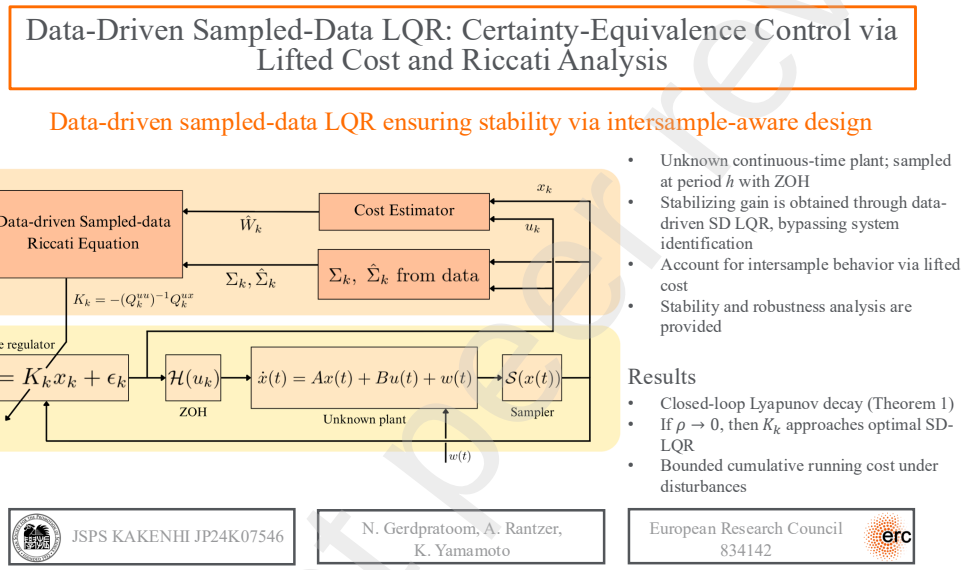
Graphical Abstract

Data-Driven Sampled-Data LQR: Certainty-Equivalence Control via Lifted Cost and Riccati Analysis

Nuthasith Gerdpratoom, Anders Rantzer, Kaoru Yamamoto



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Highlights

Data-Driven Sampled-Data LQR: Certainty-Equivalence Control via Lifted Cost and Riccati Analysis

Nuthasith Gerdpratoom, Anders Rantzer, Kaoru Yamamoto

- Proposes a Riccati-based framework to compute sampled-data LQR gains directly from input–state trajectories, without system identification.
- Accurately captures intersample dynamics by formulating the LQR cost over lifted (continuous-time) behavior.
- Ensures closed-loop stability and robustness comparable to standard model-based sampled-data LQR under mild assumptions.
- Introduces a practical algorithm using semidefinite programming and streaming data for real-time controller updates.
- Data-driven method to incorporate intersample cost directly, bridging data-driven control with sampled-data theory.

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Data-Driven Sampled-Data LQR: Certainty-Equivalence Control via Lifted Cost and Riccati Analysis

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Abstract

A data-driven design framework for sampled-data LQR is presented, in which the continuous-time plant is regulated by a digital controller with a zero-order hold when its model is unknown. This paper extends Rantzer's recent work on a data-driven Riccati equation to a sampled-data setting. A data-driven sampled-data Riccati equation is derived, from which its result can be obtained directly from input–state trajectories collected at the sampling instants. The conditions for stability and robustness of the closed-loop system are derived subsequently, adapting the analysis to account for intersample behavior via lifted cost formulations. It is demonstrated that, once a sufficiently large set of data trajectories has been collected, the closed-loop system attains stability and robustness comparable to those of the standard sampled-data LQR with full model knowledge.

Keywords: Data-driven control, sampled-data systems, intersample dynamics, certainty-equivalence control

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1. Introduction

In this letter, an unknown continuous-time linear system controlled by a digital computer with periodic sampling and zero-order hold actuation is considered. The goal is to attain LQR-optimal control using measured data instead of an explicitly known model. This model-free formulation aligns with the growing trend of data-driven control, where controllers regulating unknown systems are synthesized from input–output or state trajectory data. The recent work on a data-driven Riccati equation [1] and Linear Quadratic Dual Control [2] provides a foundation for this work, where a certainty-equivalence adaptive controller is analyzed via a data-driven Riccati equation with the correlation matrices that encode input-output data collected along the trajectory, analogous to the Bellman optimality equation in Q-learning. The result is that the adaptive controller achieves stability and is robust to unmodeled dynamics. This work is built on the same idea but adapted to sampled-data LQR, where the digital controller must be able to address the continuous-time plant with intermittent control updates. This setting introduces challenges since the intersample behavior can degrade performance or stability if not properly handled. Recent research continues to refine optimal sampled-data control strategies, e.g. [3] added an impulsive term alongside the standard control input to reduce conservativeness in choosing the sampling period. There is a modern treatment for sampled-data systems called *lifting*, which allows sampled-data systems to be represented as time-invariant models while preserving intersample behavior [4, 5, 6], and [7].

Since data-driven strategies have become a fast-growing trend in the con-

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trol community [8], numerous studies have focused on data-driven or adaptive control approaches to LQR, which involve computing the feedback gain from collected data without system identification. For instance, [9] presented a method to compute the infinite-horizon gain using Markov parameters with input-state data. [10] proposed a closed-form estimate of the LQR gain based on collected data. [11] derived explicit formulas for data-driven control that ensure stabilization and optimality, bridging Willems' fundamental lemma [12] with the LQR problem. [13] proved that a simple certainty-equivalence strategy attains nearly optimal performance for linear systems. In parallel, the reinforcement learning community has also addressed LQR, as seen in [14, 15, 16], and [17]. To the best of our knowledge, a data-driven approach to LQR that considers intersample dynamics in the cost has not yet been explored and will be presented here. It may seem straightforward to obtain an optimal gain indirectly by estimating a continuous-time model and then solving the standard discrete-time Riccati equation with coupling terms. Nonetheless, this introduces numerical instabilities. Thus, in this paper, a direct method that bypasses system identification is addressed. Likewise, we analyze the robustness of the closed-loop system based on the worst-case model representations of disturbances, building on the formulations in [18, 19], and [20]

The contribution of this paper is divided into two-fold. First, we propose a direct data-driven approach to the sampled-data LQR problem through solving the data-driven sampled-data Riccati equation with cost estimation via semidefinite programming. In addition, we present an on-line data-driven sampled-data control algorithm for practical deployment. Second, we quan-

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tify the difference between the proposed method and a sampled-data LQR when the continuous-time system matrices are fully known. The theorem utilized to guarantee stability and robustness of the closed-loop system is developed similarly to [1, 2].

The paper is outlined as follows. Section 2, preliminaries, recaps continuous-time LQR and the lifted formulation for sampled-data systems. In Section 3, a data-driven sampled-data Riccati equation is introduced as well as an on-line data-driven LQR algorithm. Section 4 formulates the problem. We then present our main result in Section 5, and then conclude the paper in Section 6. The proof of the main theorem and corollary is given in the appendix.

2. Preliminaries

2.1. Standard Continuous-time LQR

Consider the continuous-time linear system

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x(0) = x_0, \tag{1}$$

with state $x \in \mathbb{R}^n$, input $u \in \mathbb{R}^m$. $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$ are system matrices, where the pair (A, B) is assumed to be stabilizable. An infinite-horizon continuous-time LQR cost function is

$$J = \int_0^\infty \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}^\top \begin{bmatrix} W_x & \\ & W_u \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} \mathrm{d}t, \tag{2}$$

where $W_x \succ 0$ and $W_u \succ 0$ are weight matrices corresponding to state x and control effort u, respectively. The goal is to find a control input $u : [0, \infty) \to \mathbb{R}^m$ that minimizes the objective function (2). Suppose that the

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control input takes the state feedback form $u(t) = Kx(t)$, the controller gain K can be obtained by solving an algebraic Riccati equation (ARE) below off-line [21, 22]:

$$A^\top P + PA - PBW_u^{-1}B^\top P + W_x = 0, \tag{3}$$

where $P \in \mathbb{S}^n_{++}$ is a unique solution matrix, and the state feedback gain is $K = -W_u^{-1}B^\top P$. Note that $\mathbb{S}^n_{++}$ denotes the set of $n \times n$ real symmetric positive definite matrices.

2.2. Sampled-data LQR via Lifted Cost

In an actual implementation of LQR, an analogue plant is regulated by a digital controller. The plant's state $x(t)$ is sampled, and the control input $u(t)$ is held constant at the interface between discrete-time and continuous-time components. Such a system is referred to as a sampled data system. The sampling signal of the system's state is described as:

$$x_k = x(kh), \tag{4}$$

where $h \in \mathbb{R}_+$ is sampling period, and $k \in \mathbb{Z}$. In this paper, the measurements of the state $x(t)$ are assumed to be available at a high rate. At the same time, the controller updates only at multiples of the sampling period and applies a zero-order hold (ZOH) between the updates:

$$u(t) = u_k, \quad kh \leq t < kh + h. \tag{5}$$

To precisely handle the LQR problem in sampled-data control settings, sampling and ZOH must be taken into account.

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Consider a solution to the linear time-invariant state equation (1) parametrized by $\theta \in [0, h)$ below:

$$x(kh+\theta) = e^{A\theta} x_k + \int_0^{\theta} e^{A\tau} B \, \mathrm{d}\tau u_k,$$

where x_k defined in (4). The intersample solution presented above can be expressed in a more concise form:

$$\begin{bmatrix} x(kh+\theta) \\ u(kh+\theta) \end{bmatrix} = \begin{bmatrix} e^{A\theta} & \int_0^{\theta} e^{A\tau B \mathrm{d}\tau} \\ 0 & I \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix} = e^{M\theta} \begin{bmatrix} x_k \\ u_k \end{bmatrix},$$

where

$$M = \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix} \quad \text{(Lemma 10.5.1, [23])}.$$

This leads to the following discrete-time cost function equivalent to (2):

$$\begin{aligned} J &= \sum_{k=0}^{\infty} \int_{kh}^{kh+h} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}^{\top} \begin{bmatrix} W_x & \\ & W_u \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} \mathrm{d}t \\ &= \sum_{k=0}^{\infty} \int_0^{h} \begin{bmatrix} x(kh+\theta) \\ u(kh+\theta) \end{bmatrix}^{\top} \begin{bmatrix} W_x & \\ & W_u \end{bmatrix} \begin{bmatrix} x(kh+\theta) \\ u(kh+\theta) \end{bmatrix} \mathrm{d}\theta \\ &= \sum_{k=0}^{\infty} \begin{bmatrix} x_k \\ u_k \end{bmatrix}^{\top} \int_0^{h} e^{M^{\top}\theta} \begin{bmatrix} W_x & \\ & W_u \end{bmatrix} e^{M\theta} \, \mathrm{d}\theta \begin{bmatrix} x_k \\ u_k \end{bmatrix} \\ &:= \sum_{k=0}^{\infty} \begin{bmatrix} x_k \\ u_k \end{bmatrix}^{\top} \bar{W} \begin{bmatrix} x_k \\ u_k \end{bmatrix}, \end{aligned} \tag{6}$$

where the lifted weighted matrix $\bar{W} \in \mathbb{S}_{++}^{n+m}$ is defined as:

$$\bar{W} := \begin{bmatrix} \bar{W}_{xx} & \bar{W}_{ux}^{\top} \\ \bar{W}_{ux} & \bar{W}_{uu} \end{bmatrix} = \int_0^{h} e^{M^{\top}\theta} \begin{bmatrix} W_x & \\ & W_u \end{bmatrix} e^{M\theta} \, \mathrm{d}\theta.$$

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The sampled-data LQR problem is then formulated as follows:

$$\begin{aligned} \underset{u_k \in \mathbb{R}^m,\, k \in \mathbb{N} \cup \{0\}}{\text{Minimize}} \quad & \sum_{k=0}^{\infty} \begin{bmatrix} x_k \\ u_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} x_k \\ u_k \end{bmatrix} \\ \text{subject to} \quad & x_{k+1} = A_\mathrm{d} x_k + B_\mathrm{d} u_k, \end{aligned} \tag{7}$$

with $x_0 \in \mathbb{R}^n$ given, and A_d and B_d can be obtained through step-invariant transform (Theorem 3.1.1, [23]):

$$A_\mathrm{d} := e^{Ah}, \quad B_\mathrm{d} := \int_0^h e^{A\tau} B \mathrm{d}\tau.$$

Since the pair (A, B) is assumed to be stabilizable, so is $(A_\mathrm{d}, B_\mathrm{d})$. The state-feedback gain K can be obtained by solving the discrete-time ARE in a minimization form:

$$P = \min_K \left(\begin{bmatrix} I \\ K \end{bmatrix}^\top \left(\bar{W} + [A_\mathrm{d} \quad B_\mathrm{d}]^\top \; P \; [A_\mathrm{d} \quad B_\mathrm{d}] \right) \begin{bmatrix} I \\ K \end{bmatrix} \right) \tag{8}$$

$$\begin{aligned} &= \min_K \left(\bar{W}_{xx} + K^\top \bar{W}_{uu} K + 2K^\top \bar{W}_{ux} \right. \\ &\quad \left. + (A_\mathrm{d} + B_\mathrm{d} K)^\top P (A_\mathrm{d} + B_\mathrm{d} K) \right), \end{aligned} \tag{9}$$

where $P \in \mathbb{S}^n_{++}$ is a unique solution matrix to the discrete-time Riccati equation (9), and the minimizing argument yields the optimal feedback gain K [24]:

$$K = -(B_\mathrm{d}^\top P B_\mathrm{d} + \bar{W}_{uu})^{-1} (B_\mathrm{d}^\top P A_\mathrm{d} + \bar{W}_{ux}).$$

In practice, for sampled-data systems with ZOH input, once continuous-time system matrices A and B are known, and weighted matrices W_x and W_u are specified, an optimal feedback gain K minimizing the cost function (6) can be obtained by employing Riccati solvers, such as `idare` function in

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MATLAB with coupling penalty terms, where the lifted weighted matrix $\bar{W}$ can be computed through a numerical integration.

3. Data-Driven Sampled-Data LQR and Cost Estimation

In model-free LQR, where the continuous-time matrices A and B are unavailable, a common workaround is to (i) estimate the discrete-time system matrics pair $(A_\mathrm{d}, B_\mathrm{d})$ from input–state trajectories through least-squares, (ii) apply the inverse step-invariant transform to recover the continuous-time dynamics (A, B), which is needed to compute $\bar{W}$, and (iii) solve the discrete-time ARE (8) at every time step to obtain the time-varying policy $u_k = K_k x_k$. However, the procedure requires computing a matrix logarithm and a matrix inverse, both of which are numerically sensitive operations. In this article, we address the direct method, which bypasses model identification and learn the control policy $u(x_k)$ directly from data. The resulting policy is optimal with respect to the lifted cost function (6) that reflects regulating sampled-data systems.

Following the derivation in [2] and [1], define the augmented cost matrix $Q := \bar{W} + [A_\mathrm{d} \quad B_\mathrm{d}]^\top P\, [A_\mathrm{d} \quad B_\mathrm{d}] \in \mathbb{R}^{(n+m)\times(n+m)}$. Then, the discrete-time ARE (8) becomes

$$Q - \bar{W} = [A_\mathrm{d} \quad B_\mathrm{d}]^\top \min_K \left(\begin{bmatrix} I \\ K \end{bmatrix}^\top Q \begin{bmatrix} I \\ K \end{bmatrix} \right) [A_\mathrm{d} \quad B_\mathrm{d}]\,. \tag{10}$$

Equivalently, for any state-input (x, u), we obtain an expression similar to

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the model-free form in [2] and [1] below:

$$\begin{bmatrix} x \\ u \end{bmatrix}^\top \left(Q - \bar{W}\right) \begin{bmatrix} x \\ u \end{bmatrix} = x_+^\top \min_K \left(\begin{bmatrix} I \\ K \end{bmatrix}^\top Q \begin{bmatrix} I \\ K \end{bmatrix} \right) x_+, \tag{11}$$

where $x_+ = A_\mathrm{d} x + B_\mathrm{d} u$. Note, however, that (11) is not a model-free expression as a continuous-time system matrix pair has to be known to compute $\bar{W}$ analytically. The estimation of $\bar{W}$, bypassing the system identification, is discussed later in this section. Consequently, one can estimate Q directly from data tuples (x, u, x_+). Suppose that the state-input trajectory data $x_0, u_0, x_1, u_1, \ldots, x_{k-1}, u_{k-1}, x_k$ is available. Stacking these data samples in (11) gives

$$\begin{aligned} &\begin{bmatrix} x_0 & \ldots & x_{k-1} \\ u_0 & \ldots & u_{k-1} \end{bmatrix}^\top \left(Q - \bar{W}\right) \begin{bmatrix} x_0 & \ldots & x_{k-1} \\ u_0 & \ldots & u_{k-1} \end{bmatrix} \\ &= \begin{bmatrix} x_1 & \ldots & x_k \end{bmatrix}^\top \min_K \left(\begin{bmatrix} I \\ K \end{bmatrix}^\top Q \begin{bmatrix} I \\ K \end{bmatrix} \right) \begin{bmatrix} x_1 & \ldots & x_k \end{bmatrix}. \end{aligned} \tag{12}$$

Pre- and post-multiplying by

$$\begin{bmatrix} \lambda^k x_0 & \lambda^{k-1} x_1 & \ldots x_{k-1} \\ \lambda^k u_0 & \lambda^{k-1} u_1 & \ldots u_{k-1} \end{bmatrix},$$

where $\lambda \in (0, 1)$ is the forgetting factor implemented to emphasize recent samples while still exploiting the entire history, yields a *data-driven sampled-data Riccati equation*

$$\Sigma_k (Q_k - \bar{W}) \Sigma_k = \hat{\Sigma}_k^\top \min_K \left(\begin{bmatrix} I \\ K \end{bmatrix}^\top Q_k \begin{bmatrix} I \\ K \end{bmatrix} \right) \hat{\Sigma}_k, \tag{13}$$

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where Σ_k and $\hat{\Sigma}_k$ are correlation matrices, which encode the trajectory data collected, defined as

$$\Sigma_k = \sum_{\ell=0}^{k-1} \lambda^{k-1-\ell} \begin{bmatrix} x_\ell \\ u_\ell \end{bmatrix} \begin{bmatrix} x_\ell \\ u_\ell \end{bmatrix}^\top + \lambda^k \Sigma_0, \tag{14}$$

and

$$\hat{\Sigma}_k = \sum_{\ell=0}^{k-1} \lambda^{k-1-\ell} x_{\ell+1} \begin{bmatrix} x_\ell \\ u_\ell \end{bmatrix}^\top, \tag{15}$$

respectively, where $\lambda^\ell \Sigma_0$ is a regularization term added to make Σ_k invertible.

Note that (13) still requires knowledge of the weight matrix $\bar{W}$, which depends on the continuous-time dynamics A and B. Therefore, the matrix $\bar{W}$ needs to be estimated.

From the lifted cost function (6), let us define a running cost as follows:

$$\begin{aligned} J_k &:= \begin{bmatrix} x_k \\ u_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} x_k \\ u_k \end{bmatrix} \\ &= \int_{kh}^{kh+h} \begin{bmatrix} x(t) \\ u_k \end{bmatrix}^\top \begin{bmatrix} W_x & \\ & W_u \end{bmatrix} \begin{bmatrix} x(t) \\ u_k \end{bmatrix} \mathrm{d}t. \end{aligned} \tag{16}$$

Assuming that high-rate samples of the state $x(t)$ are available, the inter-sample cost J_k over the interval $[kh, kh+h)$ can be approximated using a numerical integration method. Then, the matrix $\bar{W}$ can be estimated by solving the following semidefinite programming (SDP) every time step k in a moving horizon fashion to limit the problem's size:

$$\begin{aligned} \underset{\hat{W}_k}{\text{Minimize}} \quad & \sum_{\ell=\max\{0,k-L\}}^{k-1} \left(\begin{bmatrix} x_\ell \\ u_\ell \end{bmatrix}^\top \hat{W}_k \begin{bmatrix} x_\ell \\ u_\ell \end{bmatrix} - J_k \right)^2 \\ \text{subject to} \quad & \hat{W}_k \succeq 0, \end{aligned} \tag{17}$$

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where $\hat{W}_k$ is the data-driven estimate of $\bar{W}$, and $L \in \mathbb{N}$ is a horizon length, which is sufficiently large.

Once $\bar{W}$ has been estimated, the data-driven sampled-data Riccati equation (13) can be solved similarly to a standard discrete-time ARE, yielding the resulting cost matrix

$$Q = \begin{bmatrix} Q^{xx} & Q^{xu} \\ Q^{ux} & Q^{uu} \end{bmatrix} \succ 0,$$

where $Q^{xx} \in \mathbb{R}^{n\times n}$, $Q^{xu} = (Q^{ux})^\top \in \mathbb{R}^{n\times m}$, and $Q^{uu} \in \mathbb{R}^{m\times m}$. The controller gain K_k that renders a greedy control policy is computed from the following expression:

$$\min_K \begin{bmatrix} I \\ K \end{bmatrix}^\top Q \begin{bmatrix} I \\ K \end{bmatrix} = Q^{xx} - Q^{xu}(Q^{uu})^{-1}Q^{ux},$$

and the minimizing argument gives the gain

$$K_k \;=\; -(Q_k^{uu})^{-1}Q_k^{ux}. \tag{18}$$

The control policy is defined as follows:

$$u(t) = K_k x_k + \epsilon_k, \quad kh \le t < kh + h, \tag{19}$$

where ϵ_k is a small, zero-mean excitation term added into the control input to provide exploration while the policy exploits its current estimate, the overall process is summarized in Algorithm 1 below.

4. Adaptive Control Architecture and Setup

Consider a continuous-time linear time-invariant plant with an uncertainty $w(t)$ modeled as white noise:

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t), \quad t \in \mathbb{R}_+ \tag{20}$$

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Algorithm 1 On-line Data-driven Sampled-data LQR

Input: sampling period h, initial state x_0, weighted matrices W_x and W_u, horizon length L, regularization matrix Σ_0, forgetting factor $\lambda \in (0, 1)$
Initialize controller gain K_0
$u_0 \leftarrow K_0 x_0 + \epsilon_0$
for $k = 1, 2, \ldots$ **do**
    Apply u_{k-1} with ZOH and measure x_k
    Store tuple (x_{k-1}, u_{k-1}, x_k)
    Compute running-cost estimate J_k through (16)
    Solve SDP (17) to get $\hat{W}_k$
    Compute correlation matrices Σ_k (14) and $\hat{\Sigma}_k$ (15)
    Solve Riccati Eq. (13) to obtain Q_k
    $K_k \leftarrow -(Q_k^{uu})^{-1} Q_k^{ux}$ ▷ see (18)
    $u_k \leftarrow K_k x_k + \epsilon_k$ ▷ see (19)
end for

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and let the state be sampled every sampling period $h > 0$ as in (4), which is $x_k := x(kh)$ for $k \in \mathbb{Z}$. We will analyze controllers of the following form:

$$\begin{cases} \Sigma_k &= \lambda \Sigma_{k-1} + \left[x_{k-1}^\top u_{k-1}^\top\right]^\top \left[x_{k-1}^\top u_{k-1}^\top\right], \\ \hat{\Sigma}_k &= \lambda \hat{\Sigma}_{k-1} + x_k \left[x_{k-1}^\top u_{k-1}^\top\right], \\ u(t) &= K_k x_k + \epsilon_k, \quad kh \le t < kh + h \end{cases} \tag{21}$$

where $\lambda \in (0, 1)$. At the end of every time step, the feedback gain matrix K_k is updated by solving the data–driven sampled-data LQR problem in (13) with the current correlation matrices Σ_k and $\hat{\Sigma}_k$, while the lifted weighted matrix $\bar{W}$ is obtained by solving the SDP (17).

We will establish similar conditions as in [1] and [2] that guarantee stability and performance in terms of the continuous-time cost function J defined in (2). For this, we pose the following prior constraints: *Given any $\beta > 1$, model parameters are restricted to be in the set $\mathcal{M}_\beta$ consisting of all pairs $(A_\mathrm{d}, B_\mathrm{d})$ such that the Riccati equation* (10) *has a solution Q with $\bar{W} \preceq Q \preceq \beta^2 \bar{W}$.*

Following the preceding works, we introduce the correlation matrices between the disturbance w and the past data as follows:

$$\left[\Sigma_k^{wx} \quad \Sigma_k^{wu}\right] := \sum_{\ell=0}^{k-1} \lambda^{k-1-\ell} w_\ell \left[x_\ell^\top \quad u_\ell^\top\right] - \lambda^k \left[A_\mathrm{d} \quad B_\mathrm{d}\right] \Sigma_0,$$

where

$$w_\ell := \int_0^h e^{A(h-\tau)} w(\ell h + \tau)\, d\tau.$$

Note that $\left[\Sigma_k^{wx} \quad \Sigma_k^{wu}\right] = \hat{\Sigma}_k - \left[A_\mathrm{d} \quad B_\mathrm{d}\right] \Sigma_k$.

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5. Stability and Robustness Analysis

Theorem 1. *Consider $\beta, \rho \in \mathbb{R}_+$ and $\beta > 1$ with $2\beta^2\rho(\rho+2) < 1$, a matrix $\hat{\Sigma}_k$ and a positive definite matrix Σ_k. Let $(A_\mathrm{d}, B_\mathrm{d}) \in \mathcal{M}_\beta$ with P satisfying Riccati equation* (9)*, while*

$$\left\| [A_\mathrm{d} \quad B_\mathrm{d}] - \hat{\Sigma}_k \Sigma_k^{-1} \right\| \leq \rho. \tag{22}$$

Then (13) *has a solution with minimizer K_k and*

$$\frac{1}{1-\delta} P \succeq \begin{bmatrix} I \\ K_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ K_k \end{bmatrix} + (A_\mathrm{d} + B_\mathrm{d} K_k)^\top P (A_\mathrm{d} + B_\mathrm{d} K_k), \tag{23}$$

where $\delta := 2\beta^2\rho(\rho+2) \in \mathbb{R}_+$.

Proof. See Appendix. □

Suppose that $\bar{K}$ is an optimal gain satisfying the standard Riccati equation (9) with the solution matrix P, that is

$$P = \begin{bmatrix} I \\ \bar{K} \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ \bar{K} \end{bmatrix} + (A_\mathrm{d} + B_\mathrm{d} \bar{K})^\top P (A_\mathrm{d} + B_\mathrm{d} \bar{K}). \tag{24}$$

Let us define the Lyapunov function as $V_k := \|x_k\|_P^2$. With the optimal control policy $u_k = \bar{K} x_k$, the Lyapunov function decreases over time as follows:

$$V_{k+1} - V_k = -x_k^\top \begin{bmatrix} I \\ \bar{K} \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ \bar{K} \end{bmatrix} x_k < 0. \tag{25}$$

The Lyapunov decay presented above is a standard stability guarantee for an LQR problem, where the cost function includes coupling terms. In a data-driven sampled-data LQR setting, where the greedy policy gain K_k is

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obtained from (13), the Lyapunov decay in (25) is strictly attained once a sufficiently large dataset has been collected. Theorem 1 quantifies the difference in stability between the proposed data-driven sampled-data LQR and the conventional discrete-time LQR. With the expression (23) from the theorem, the Lyapunov inequality becomes

$$V_{k+1} - V_k \leq x_k^\top \left(\frac{\delta}{1-\delta} P - \begin{bmatrix} I \\ K_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ K_k \end{bmatrix} \right) x_k. \tag{26}$$

From the expression (26), the Lyapunov function might fail to decay initially. Nonetheless, as more data are collected, the identification error shrinks and the estimated discrete-time matrices $(\hat{A}_\mathrm{d}, \hat{B}_\mathrm{d})$ converge to $(A_\mathrm{d}, B_\mathrm{d})$ as $\rho \to 0$ and $\frac{\delta}{1-\delta}P$ term in (26) vanishes. In other words, as the difference between the left-hand side of (23) and (24) is adequately small, stability is maintained. The strict decay rate gives robustness to deviations from optimality. Note that the decay of the storage function at discrete time steps also guarantees the stability of the sampled-data system. This is used to derive the following corollary.

Corollary 1. *Suppose that* $(A_\mathrm{d}, B_\mathrm{d}) \in \mathcal{M}_\beta$ *and* $P \in \mathbb{S}^n_{++}$ *is a unique solution matrix to the Riccati equation* (8)*. Consider the closed-loop dynamics*

$$x_{k+1} = A_\mathrm{d} x_k + B_\mathrm{d} u_k + w_k,$$

$$u_k = K_k x_k + \epsilon_k,$$

and suppose that $\Sigma_k, \hat{\Sigma}_k$ *are defined by* (14) *and* (15)*, the inequality* (22) *holds for all* $k \geq k_0$*. For each value of* k*, let* Q_k *be a solution to the equation* (13) *that minimizes the argument* K_k*. Let us define*

$$\alpha := \beta^2 + \frac{1}{1-\beta^2/\gamma^2} \left(1 - \frac{\beta^2}{1 - 2\beta^2\rho(\rho+2)} \right).$$

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If $\alpha > 0$, then

$$\sum_{k=k_0}^{T-1} x_k^\top \begin{bmatrix} I \\ K_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ K_k \end{bmatrix} x_k \\ \leq \frac{1}{\alpha}\|x_{k_0}\|_P^2 + \frac{\gamma^2}{\alpha}\sum_{k=k_0}^{T-1}\|B\epsilon_k + w_k\|_{\bar{W}}^2. \tag{27}$$

Thus, the cumulative running cost is upper-bounded.

Proof. See Appendix. □

6. Conclusion

In this letter, we proposed a data-driven adaptive control framework for sampled-data systems that directly addresses continuous-time performance through a lifted cost formulation. Building on the data-driven Riccati analysis in [1, 2], we developed a certainty-equivalence adaptive controller that estimates both the lifted cost matrix and optimal policy from trajectory data, without system identification. Our main result extends Riccati-based stability guarantees to the sampled-data setting under suitable assumptions. An online implementation scheme was introduced that combines cost estimation via semidefinite programming with policy updates from data. Future work will investigate robustness to cost-estimation errors and provide a more detailed analysis of intersample behavior and performance degradation.

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Appendix

Lemma 1. *Consider positive numbers β and ρ, matrices $\hat{\Sigma}, \tilde{\Sigma}$ and positive semi-definite matrices Σ, P, Q. Suppose that $(\hat{\Sigma}-\tilde{\Sigma})^\top P(\hat{\Sigma}-\tilde{\Sigma}) = \Sigma(Q-\bar{W})\Sigma$ and $\tilde{\Sigma}^\top \bar{W}\tilde{\Sigma} \preceq \rho^2\Sigma\bar{W}\Sigma$. If $\bar{W} \preceq Q \preceq \beta^2\bar{W}$, then $\hat{\Sigma}^\top P\hat{\Sigma} \preceq \Sigma Q\Sigma + (\beta^2\rho(\rho+2)-1)\Sigma\bar{W}\Sigma$.*

Proof. For every z with appropriate dimension,

$$\begin{aligned}
\|\hat{\Sigma}z\|_P^2 &\leq \Big(\|(\hat{\Sigma}-\tilde{\Sigma})z\|_P + \|\tilde{\Sigma}z\|_P\Big)^2 \\
&= \Big(\|\Sigma z\|_{Q-\bar{W}} + \|\tilde{\Sigma}z\|_P\Big)^2 \\
&\leq \big(\|\Sigma z\|_{Q-\bar{W}} + \beta\rho\|\Sigma z\|_{\bar{W}}\big)^2 \\
&= \|\Sigma z\|_{Q-\bar{W}}^2 + \beta^2\rho^2\|\Sigma z\|_{\bar{W}}^2 + 2\|\Sigma z\|_{Q-\bar{W}}\beta\rho\|\Sigma z\|_{\bar{W}}^2 \\
&\leq \|\Sigma z\|_{Q-\bar{W}}^2 + \beta^2\rho^2\|\Sigma z\|_{\bar{W}}^2 + 2\beta^2\rho\|\Sigma z\|_{\bar{W}}^2 \\
&= \|\Sigma z\|_Q^2 + (\beta^2\rho(\rho+2)-1)\|\Sigma z\|_{\bar{W}}^2.
\end{aligned}$$

The first inequality is the triangle inequality, and the second inequality follows from $P \preceq \beta^2\bar{W}$ and $\tilde{\Sigma}^\top\bar{W}\tilde{\Sigma} \preceq \rho^2\Sigma\bar{W}\Sigma$. The third inequality follows from the assumption $\bar{W} \preceq Q \preceq \beta^2\bar{W}$. This completes the proof. □

Lemma 2. *Consider matrices $A_\mathrm{d}, B_\mathrm{d}, \bar{K}$, and $\bar{Q} \succeq \bar{W}$. Suppose*

$$[A_\mathrm{d} \quad B_\mathrm{d}]^\top \begin{bmatrix} I \\ \bar{K} \end{bmatrix}^\top \bar{Q} \begin{bmatrix} I \\ \bar{K} \end{bmatrix} [A_\mathrm{d} \quad B_\mathrm{d}] \preceq \bar{Q} - \bar{W}. \tag{28}$$

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Then there exists a unique Q with $\bar{W} \preceq Q \preceq \bar{Q}$ such that

$$[A_\mathrm{d} \quad B_\mathrm{d}]^\top \min_K \left(\begin{bmatrix} I \\ K \end{bmatrix}^\top Q \begin{bmatrix} I \\ K \end{bmatrix} \right) [A_\mathrm{d} \quad B_\mathrm{d}] = Q - \bar{W}. \tag{29}$$

Proof. The matrix

$$\bar{P} := \begin{bmatrix} I \\ \bar{K} \end{bmatrix}^\top \bar{Q} \begin{bmatrix} I \\ \bar{K} \end{bmatrix}$$

satisfies

$$\begin{aligned} \bar{P} =& (A_\mathrm{d} + B_\mathrm{d}\bar{K})^\top \bar{P}(A_\mathrm{d} + B_\mathrm{d}\bar{K}) + \bar{W}_{xx} \\ &+ \bar{K}^\top \bar{W}_{uu}\bar{K} + 2\bar{K}^\top \bar{W}_{ux}. \end{aligned}$$

Therefore, $P_0 = \bar{P}$ and the Riccati iteration

$$\begin{aligned} P_{k+1} = \min_K ((&A_\mathrm{d} + B_\mathrm{d}\bar{K})^\top P_k(A_\mathrm{d} + B_\mathrm{d}\bar{K}) + \bar{W}_{xx} \\ &+ \bar{K}^\top \bar{W}_{uu}\bar{K} + 2\bar{K}^\top \bar{W}_{ux}) \end{aligned}$$

gives a decreasing sequence with limit $P_\infty \succeq \bar{W}_{xx}$. Setting

$$Q := [A_\mathrm{d} \quad B_\mathrm{d}]^\top \; P_\infty \; [A_\mathrm{d} \quad B_\mathrm{d}]$$

proves the existence. Since stabilizing solutions to the Riccati equation are unique [25, Ch. 3], Q must be unique. □

Proof of Theorem 1. By definition, the assumption $(A_\mathrm{d}, B_\mathrm{d}) \in \mathcal{M}_\beta$ implies existence of Q satisfying (10) and $\bar{W} \preceq Q \preceq \beta^2\bar{W}$. Let $\bar{K}$ be the corresponding minimizing argument in (10). Define

$$\begin{aligned} P &:= \begin{bmatrix} I & \bar{K}^\top \end{bmatrix} \; Q \; \begin{bmatrix} I & \bar{K}^\top \end{bmatrix}^\top \\ \tilde{\Sigma} &= \hat{\Sigma}_k - [A_\mathrm{d} \quad B_\mathrm{d}]\, \Sigma_k. \end{aligned}$$

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Then (10) gives $(\hat{\Sigma}_k - \tilde{\Sigma}_k)^\top P(\hat{\Sigma}_k - \tilde{\Sigma}_k) = \Sigma_k(Q - \bar{W})\Sigma_k$, so Lemma 1 implies

$$\hat{\Sigma}_k^\top P \hat{\Sigma}_k \preceq \Sigma_k(Q - \check{\alpha}\bar{W})\Sigma_k,$$

where $\check{\alpha} \coloneqq 1 - \beta^2\rho(\rho + 2)$. It follows that (28) holds with $\bar{Q} = \check{\alpha}^{-1}Q$, so Lemma 2 proves existence of Q_k, and consequently K_k, satisfying (13) and $\bar{W} \preceq Q_k \preceq \check{\alpha}^{-1}Q$.

Next, define

$$\begin{aligned} P_k &\coloneqq \begin{bmatrix} I & K_k^\top \end{bmatrix} \; Q_k \; \begin{bmatrix} I & K_k^\top \end{bmatrix}^\top \\ \bar{\Sigma}_k &\coloneqq [A_\mathrm{d} \quad B_\mathrm{d}]\, \Sigma_k. \end{aligned}$$

Then the equality

$$(\bar{\Sigma}_k + \tilde{\Sigma}_k)^\top P_k(\bar{\Sigma}_k + \tilde{\Sigma}_k) = \hat{\Sigma}^\top P_k \hat{\Sigma} = \Sigma_k(Q_k - \bar{W})\Sigma_k$$

and the inequality

$$Q_k \preceq \check{\alpha}^{-1}\beta^2\bar{W}$$

allow application of Lemma 2 to get

$$\bar{\Sigma}_k^\top P_k \bar{\Sigma}_k \preceq \Sigma_k(Q_k - \hat{\alpha}\bar{W})\Sigma_k,$$

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where $\hat{\alpha} \coloneqq 1 - \check{\alpha}^{-1}\beta^2\rho(\rho+2)$. It follows from Lemma 1 that $Q \preceq \hat{\alpha}^{-1}Q$, so

$$
\begin{aligned}
&(A_\mathrm{d} + B_\mathrm{d}K_k)^\top P(A_\mathrm{d} + B_\mathrm{d}K_k) \\
&= \begin{bmatrix} I \\ K_k \end{bmatrix}^\top \begin{bmatrix} A_\mathrm{d} & B_\mathrm{d} \end{bmatrix}^\top \begin{bmatrix} I \\ K_k \end{bmatrix}^\top Q \begin{bmatrix} I \\ K_k \end{bmatrix} \begin{bmatrix} A_\mathrm{d} & B_\mathrm{d} \end{bmatrix} \begin{bmatrix} I \\ K_k \end{bmatrix} \\
&= \begin{bmatrix} I \\ K_k \end{bmatrix}^\top (Q - \bar{W}) \begin{bmatrix} I \\ K_k \end{bmatrix} \\
&\preceq \begin{bmatrix} I \\ K_k \end{bmatrix}^\top (\hat{\alpha}^{-1}Q_k - \bar{W}) \begin{bmatrix} I \\ K_k \end{bmatrix} \\
&= \hat{\alpha}^{-1}P_k - \begin{bmatrix} I \\ K_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ K_k \end{bmatrix} \\
&\preceq \hat{\alpha}^{-1}\check{\alpha}^{-1}P - \begin{bmatrix} I \\ K_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ K_k \end{bmatrix} \\
&= \big(1 - 2\beta^2\rho(\rho+2)\big)^{-1}P - \begin{bmatrix} I \\ K_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} I \\ K_k \end{bmatrix},
\end{aligned}
$$

and the proof is complete. $\square$

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Proof of Corollary 1.

$$
\begin{aligned}
&\max_{w_k} \left(\|x_{k+1}\|_P^2 - \gamma \|B\epsilon_k + w_k\|_{\bar{W}}^2 \right) \\
&= \max_{w_k} \left(\|(A_\mathrm{d} + B_\mathrm{d} K_k) x_k + w_k\|_P^2 - \gamma^2 \|w_k\|_{\bar{W}}^2 \right) \\
&= \|\gamma^2 (\gamma^2 W - P)^{-1} W (A_\mathrm{d} + B_\mathrm{d} K_k) x_k\|_P^2 \\
&\quad - \gamma^2 \|(\gamma^2 W - P)^{-1} P (A_\mathrm{d} + B_\mathrm{d} K_k) x_k\|_{\bar{W}}^2 \\
&= \|(A_\mathrm{d} + B_\mathrm{d} K_k) x_k\|_{P(I - \gamma^{-2} W^{-1} P)^{-1}} \\
&\leq (1 - \beta^2/\gamma^2)^{-1} \|(A_\mathrm{d} + B_\mathrm{d} K_k) x_k\|_P \\
&\leq \frac{1}{1 - \beta^2/\gamma^2} \left(\frac{\|x_k\|_P^2}{1 - 2\beta^2 \rho(\rho + 2)} - \begin{bmatrix} x_k \\ K_k x_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} x_k \\ K_k x_k \end{bmatrix} \right) \\
&\leq \|x_k\|_P^2 + \left(\frac{1}{(1 - \beta^2/\gamma^2)(1^2 \beta^2 \rho(\rho + 2)} - 1 \right) \beta^2 \|x_k\|_{\bar{W}}^2 \\
&\quad - \frac{1}{1 - \beta^2/\gamma^2} \begin{bmatrix} x_k \\ K_k x_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} x_k \\ K_k x_k \end{bmatrix} \\
&\leq \|x_k\|_P^2 - \alpha \begin{bmatrix} x_k \\ K_k x_k \end{bmatrix}^\top \bar{W} \begin{bmatrix} x_k \\ K_k x_k \end{bmatrix}.
\end{aligned}
$$

Summing over $k_0 \leq k \leq T - 1$ gives the desired result. □

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